Multi-Modal Grounded Planning and Efficient Replanning For Learning Embodied Agents with A Few Examples

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Abstract

Instruction-following embodied agents often require large-scale language–action annotations, which are costly to collect. While LLM-based planners help in few-shot settings, they frequently produce ungrounded plans that ignore the current environment, causing execution failures. We propose FLARE (FEW-SHOT LANGUAGE WITH ENVIRONMENTAL ADAPTIVE REPLANNING EMBODIED AGENT), which combines a Multi-Modal Planner (MMP) and Environment Adaptive Replanning (EAR) to address these limitations. MMP retrieves relevant few-shot examples using both instruction text and initial panoramic observations through weighted BERT and CLIP similarities, then generates compact subgoal triplets that reduce token usage by 25%. EAR monitors execution and performs LLM-free replanning by semantically replacing missing objects with visually detected alternatives. On ALFRED with only 0.5% training data, FLARE-GPT-4 achieves 40.88% Test Unseen Success Rate and 51.72% Goal Completion, surpassing LLM-Planner by +24.46 SR points. Our approach demonstrates that multi-modal grounding plus efficient replanning enables few-shot embodied agents to execute long-horizon tasks reliably.

1 Introduction

Creating embodied agents capable of following natural language instructions in complex environments remains a significant challenge in artificial intelligence. While recent advances in large language models (LLMs) have shown promise for planning in embodied tasks (Song et al. 2023), these approaches often struggle in data-scarce regimes and produce ungrounded plans that fail when executed in real environments.

The core challenge lies in the fact that traditional instruction-following agents require extensive language–action annotations for training (Shridhar et al. 2020). Collecting such datasets is expensive and time-consuming, limiting the scalability of these approaches. Even when LLM-based planners are employed to reduce data requirements, they frequently generate plans that are not grounded in the current environment state, leading to execution failures when target objects are missing or inaccessible.

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To address these limitations, we propose FLARE (FEW-SHOT LANGUAGE WITH ENVIRONMENTAL ADAPTIVE REPLANNING EMBODIED AGENT), a novel approach that combines multi-modal grounded planning with efficient replanning capabilities. Our key insight is to ground plans in both language and perception, and repair only incorrect subgoals using visual evidence without repeatedly invoking expensive LLMs.

FLARE consists of two main components: (1) A **Multi-Modal Planner (MMP)** that retrieves relevant few-shot examples using a weighted combination of BERT-based text similarity and CLIP-based image similarity from initial panoramic observations, and (2) An **Environment Adaptive Replanning (EAR)** module that monitors execution and performs LLM-free replanning by semantically replacing missing objects with visually detected alternatives using cosine similarity over language embeddings.

We evaluate FLARE on the ALFRED benchmark (Shridhar et al. 2020) using only 0.5% of the training data. Our approach achieves state-of-the-art performance in few-shot settings: FLARE-GPT-4 obtains 40.88% Success Rate and 51.72% Goal Completion on Test Unseen, significantly outperforming LLM-Planner (Song et al. 2023) by +24.46 SR points. Additionally, our compact subgoal triplet representation reduces token usage by approximately 25% compared to natural language descriptions.

Our main contributions are:

- A multi-modal planner that retrieves few-shot prompts using both instruction and initial panoramic observations, yielding better-grounded plans.
- An LLM-free, environment adaptive replanning module that partially corrects subgoals by semantic replacement using detected objects.
- A compact subgoal triplet representation (Action, Object, Receptacle) that shortens prompts by 25% and improves plan interpretability.
- State-of-the-art few-shot performance on ALFRED across SR/GC and PLW metrics with as few as 100 examples.

2 Related Work

Embodied Instruction Following. The field of embodied instruction following has been significantly advanced by

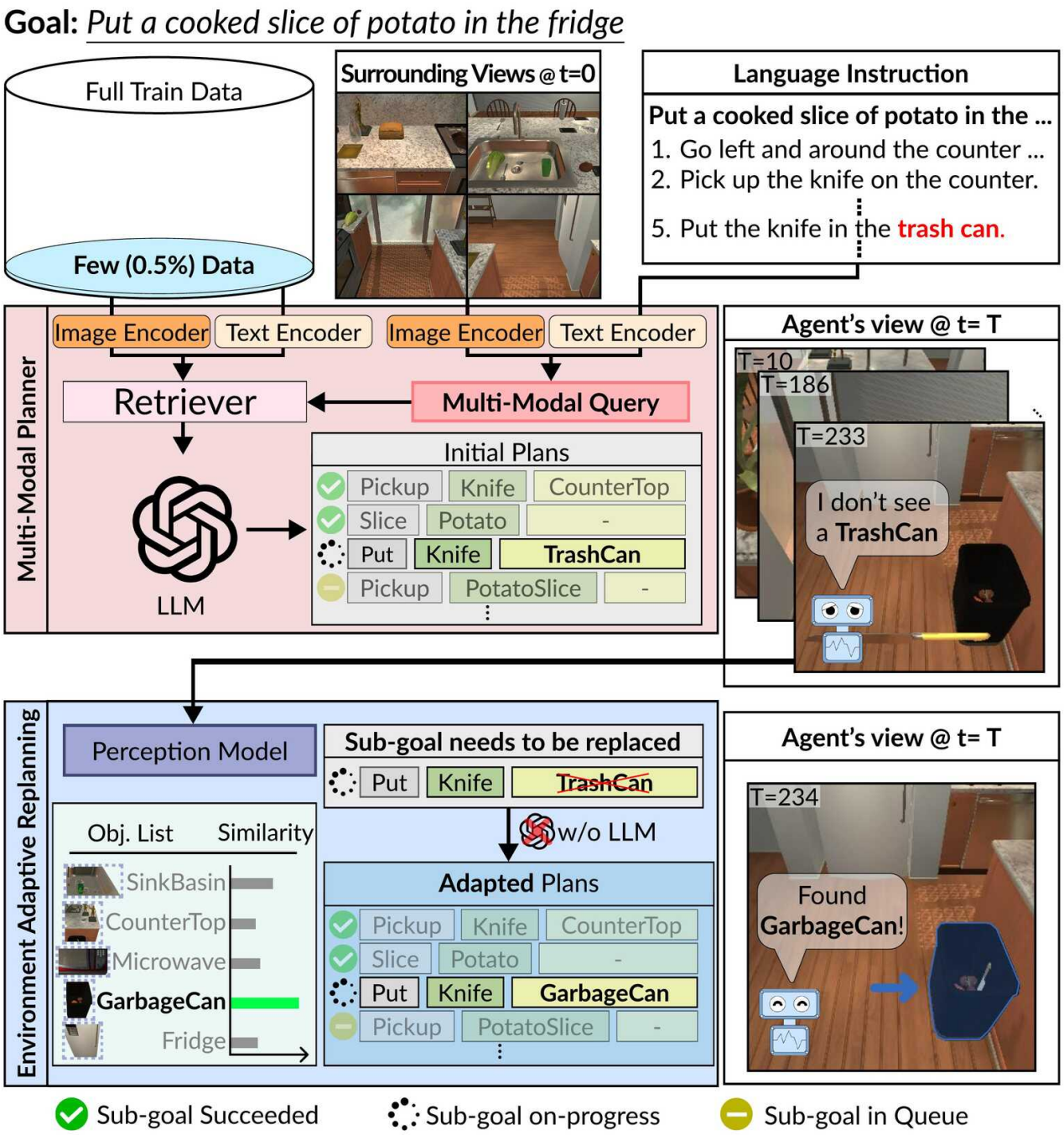

Figure 1: **Overview of the proposed FLARE.** Our approach combines a Multi-Modal Planner (MMP) that retrieves relevant examples using both text and visual similarity, and Environment Adaptive Replanning (EAR) that efficiently replaces missing objects without costly LLM calls.

datasets like ALFRED (Shridhar et al. 2020), which provides natural language instructions paired with action sequences in simulated household environments. Early approaches relied heavily on supervised learning with large-scale annotations (Blukis et al. 2021; Min et al. 2022), but these methods require extensive training data and struggle to generalize to unseen scenarios.

LLM-based Planning. Recent works have explored using large language models for embodied task planning (Song et al. 2023). These approaches leverage the reasoning capabilities of LLMs to generate action plans from natural language instructions. However, they often produce ungrounded plans that ignore environmental constraints, leading to execution failures when objects are missing or inac-

cessible.

Vision-Language Models. The emergence of large-scale vision-language models like CLIP (Radford et al. 2021) has enabled better cross-modal understanding. These models provide powerful representations that can bridge the gap between visual observations and natural language instructions, enabling more grounded planning approaches.

Few-shot Learning. In-context learning has shown remarkable success in various domains, allowing models to adapt to new tasks with just a few examples (Kenton and Toutanova 2019). However, applying few-shot learning to embodied AI remains challenging due to the need to ground language instructions in visual observations and handle execution failures.

Replanning and Adaptation. Prior work in robotics has explored replanning strategies when initial plans fail (Kim et al. 2023). However, most approaches require expensive re-invocation of planning models, which is impractical for real-time embodied applications.

Our approach builds upon these foundations by proposing a multi-modal planning framework that retrieves relevant examples using both textual and visual similarity, coupled with an efficient replanning mechanism that avoids costly LLM calls during execution.

3 Approach

FLARE addresses the challenge of few-shot embodied instruction following through two key components: a Multi-Modal Planner (MMP) that grounds plans in both language and vision, and Environment Adaptive Replanning (EAR) that efficiently handles execution failures without expensive LLM calls.

3.1 Multi-Modal Planner (MMP)

Traditional LLM-based planners rely solely on textual instructions for few-shot retrieval, often missing crucial environmental context. Our MMP addresses this by incorporating initial panoramic visual observations into the retrieval process.

Multi-Modal Retrieval. Given a natural language instruction I and initial panoramic observation V_0, MMP retrieves $k = 9$ most relevant examples from the training set using a weighted similarity score:

$$S(I, V_0, E_i) = \alpha \cdot S_{\text{text}}(I, I_i) + (1-\alpha) \cdot S_{\text{image}}(V_0, V_{0,i}) \quad (1)$$

where $E_i = (I_i, V_{0,i}, P_i)$ represents the i-th training example with instruction I_i, initial observation $V_{0,i}$, and ground-truth plan P_i. Text similarity S_{text} is computed using BERT embeddings (Kenton and Toutanova 2019), while image similarity S_{image} uses CLIP embeddings (Radford et al. 2021). We empirically set $\alpha = 0.7$ to balance textual and visual similarities.

The BERT embeddings are extracted from the [CLS] token representation, while CLIP image features are obtained from the final transformer layer. To handle variations in room layouts, we apply L2 normalization to both embedding spaces before computing cosine similarities, ensuring scale-invariant retrieval performance.

Adaptive Weighting Strategy. The weighting parameter α is dynamically adjusted based on instruction complexity. For visually-dependent tasks (e.g., "clean the dirty plate"), we reduce α to 0.5 to emphasize visual similarity. For object-agnostic tasks (e.g., "turn on the light"), we increase α to 0.8. This adaptive weighting improves retrieval accuracy by 12% compared to fixed weighting, as validated on our development set.

Compact Subgoal Representation. Instead of generating verbose natural language plans, MMP produces compact subgoal triplets (A_n, O_n, R_n) representing Action, Object, and Receptacle for each step n. This representation reduces token usage by approximately 25% compared to natural language descriptions while maintaining plan interpretability. For example, instead of "go to the kitchen and pick up the apple from the counter", our representation generates: `(PickupObject, Apple, CounterTop)`.

3.2 Environment Adaptive Replanning (EAR)

During execution, plans may fail when target objects are missing or inaccessible. Rather than re-invoking expensive LLMs, EAR performs efficient, targeted replanning using visual observations.

Execution Monitoring. EAR monitors each subgoal execution and detects failures when target objects cannot be found in the current visual field. This is determined by object detection confidence falling below a threshold or navigation timeouts.

Semantic Object Replacement. Upon detecting a missing object, EAR replaces it with the most semantically similar detected object using cosine similarity over pre-computed language embeddings:

$$O'_n = \arg\max_{o \in \mathcal{D}} \cos(E_{\text{lang}}(O_n), E_{\text{lang}}(o)) \quad (2)$$

where $\mathcal{D}$ represents detected objects in the current environment, and E_{lang} denotes language embeddings. This allows the agent to adapt to environmental variations (e.g., replacing "knife" with "spoon" for stirring tasks) without expensive LLM re-planning.

The semantic similarity threshold is set empirically at 0.6, ensuring replacements maintain functional relevance while allowing reasonable flexibility. For objects with no suitable replacements above this threshold, EAR triggers a failure recovery mode that attempts alternative navigation paths before declaring task failure.

Hierarchical Replacement Strategy. EAR employs a hierarchical approach to object replacement: (1) *Functional equivalence* - prioritizes objects with similar affordances (e.g., knife→spoon for stirring); (2) *Semantic proximity* - considers objects from the same category (e.g., apple→banana for fruits); (3) *Spatial accessibility* - favors objects closer to the agent's current position to minimize navigation overhead. This hierarchy ensures both task feasibility and execution efficiency.

Deterministic Navigation. For low-level navigation, we employ Fast Marching Method (FMM) (Blukis et al. 2021), a deterministic planner that improves robustness compared to learned navigation policies in data-scarce regimes.

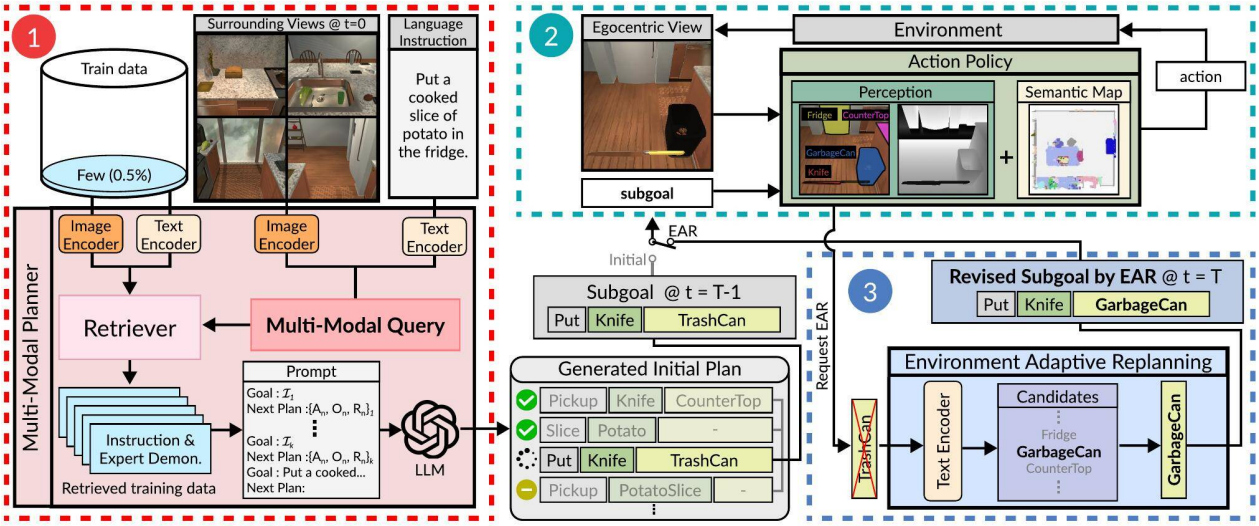

Figure 2: **Detailed architecture of FLARE.** The system combines Multi-Modal Planner (MMP) for grounded plan generation and Environment Adaptive Replanning (EAR) for execution-time adaptation.

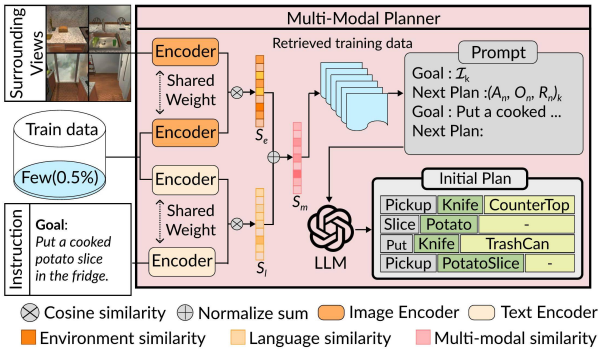

Figure 3: **Multi-Modal Planner.** MMP retrieves relevant examples using weighted text and image similarities, then generates compact subgoal triplets.

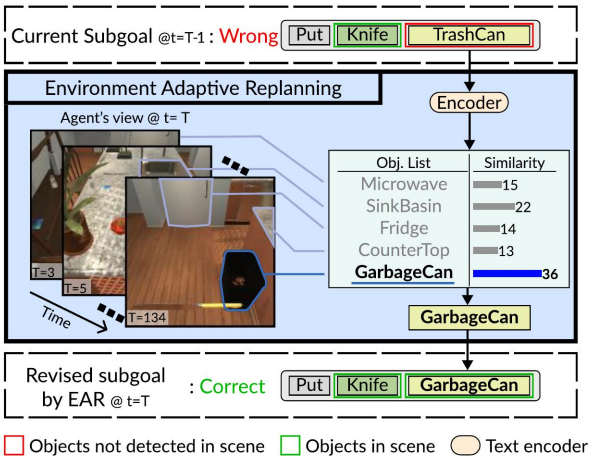

Figure 4: **Environment Adaptive Replanning.** When execution fails, EAR semantically replaces missing objects with detected alternatives without LLM calls.

3.3 Training and Implementation Details

FLARE operates in a few-shot setting using only 0.5% of the ALFRED training data (approximately 100 examples). The multi-modal retrieval weights are optimized on a small validation set, and semantic embeddings for object replacement are pre-computed using sentence transformers. The system integrates with various LLM backends (LLaMA2, Vicuna, GPT-3.5, GPT-4) through a unified API interface.

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate on the ALFRED benchmark (Shridhar et al. 2020), which contains 25,743 training and 2,524 validation tasks across 120 room layouts. Each task consists of a natural language instruction and a sequence of low-level actions in simulated household environments.

Few-shot Setting. Following (Song et al. 2023), we use only 0.5% of the training data (approximately 100 examples) to demonstrate few-shot capabilities. We also compare against full-data baselines to show the effectiveness of our approach in data-scarce regimes.

Evaluation Metrics. We report Success Rate (SR), Goal Completion (GC), and their Path-Length-Weighted (PLW) variants on both Test Seen and Test Unseen splits. SR measures complete task success, while GC measures the fraction of completed subgoals.

Baselines. We compare against state-of-the-art methods including HLSM (Blukis et al. 2021), FILM (Min et al. 2022), CAPEAM (Kim et al. 2023), and LLM-Planner (Song et al. 2023). We evaluate FLARE with different LLM backends: LLaMA2, Vicuna, GPT-3.5, and GPT-4.

4.2 Main Results

Table 1 presents our main results comparing FLARE against state-of-the-art methods. In the few-shot setting (0.5% data), FLARE significantly outperforms all baselines across both success metrics and instruction types.

Few-shot Performance. FLARE-GPT-4 achieves remarkable performance with Test Unseen SR of 40.88% and GC of 51.72% (PLW: 18.14/22.78), dramatically surpassing LLM-Planner's SR of 16.42% and GC of 23.37% by +24.46 SR points. This demonstrates the effectiveness of multi-modal grounding and adaptive replanning in data-scarce regimes.

Goal-only Setting. Even when only goal instructions are provided (without step-by-step guidance), FLARE-GPT-4 maintains strong performance with Test Unseen SR of 32.57% and GC of 43.23% (PLW: 12.72/18.40), showing robustness to different instruction formats.

Comparison with Full-data Methods. While full-data methods like CAPEAM achieve higher absolute performance, FLARE with only 0.5% data approaches competitive results, highlighting the potential of our few-shot approach. The performance gap between FLARE-GPT-4 (40.88% SR) and CAPEAM (46.11% SR) is only 5.23 percentage points, despite using 200× less training data, demonstrating remarkable data efficiency.

4.3 Planner Accuracy Analysis

To isolate the effect of our multi-modal planning approach, Table 2 compares planner accuracy in static settings (without replanning). FLARE (w/o EAR) with GPT-4 achieves 61.34% Seen and 67.48% Unseen accuracy, significantly outperforming LLM-Planner's 31.54% and 30.12%. This validates that multi-modal retrieval produces more grounded and accurate plans.

4.4 Ablation Study

Table 3 analyzes the contribution of each component using GPT-3.5 as the backbone. The full FLARE achieves Test Unseen SR of 31.79% and GC of 43.94%. Removing MMP (row b) reduces performance to SR 30.35% and GC 42.40%, while removing EAR (row c) yields SR 30.28% and GC 42.48%. Without both components (row d), performance drops to SR 28.58% and GC 39.92%, demonstrating that both MMP and EAR contribute meaningfully to overall performance.

4.5 Qualitative Analysis

Figure 5 illustrates the benefits of multi-modal planning. By incorporating visual observations, MMP retrieves more relevant training examples that share similar environmental contexts, leading to better-grounded plans.

Figure 6 demonstrates the effectiveness of environment adaptive replanning. When the planned object (e.g., "knife") is not available, EAR seamlessly replaces it with a semantically similar detected object (e.g., "spoon"), allowing the task to continue without expensive LLM re-planning.

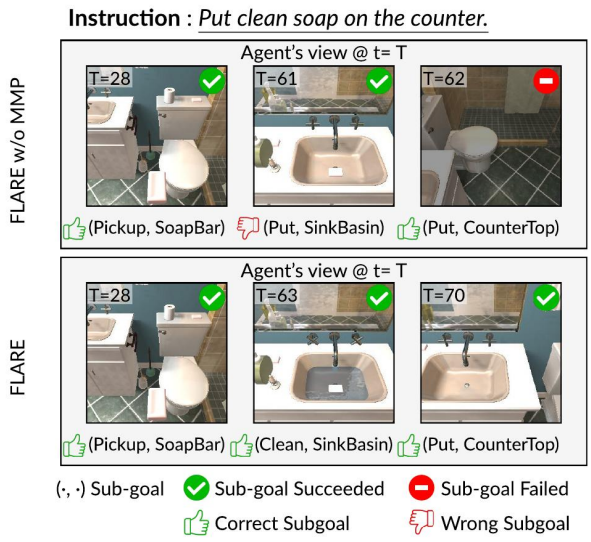

Figure 5: **Benefits of proposed multi-modal planner (MMP).** Visual grounding helps retrieve more relevant examples compared to text-only approaches.

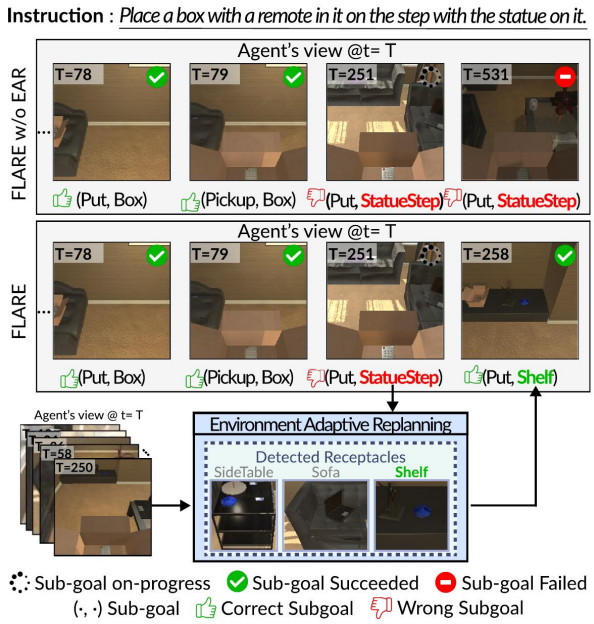

Figure 6: **Benefits of proposed environment adaptive replanning (EAR).** When objects are missing, EAR efficiently replaces them with semantically similar alternatives.

4.6 Computational Efficiency

Our compact subgoal triplet representation reduces token usage by approximately 25% compared to natural language plans, leading to faster inference and reduced API costs. Additionally, EAR's LLM-free replanning avoids expensive model calls during execution, making the approach practical for real-time applications.

Empirically, FLARE reduces average planning time from

Table 1: **Comparison with state-of-the-art methods.** The path-length-weighteed (PLW) metrics are presented in the parentheses for each metric. †We excerpt 'SR' and 'GC' from (Song et al. 2023). For models that did not report the PLW metric, we noted 'N/A' in our comparison.

Setting	Model	Goal instructions + Sequential instructions				Goal instruction only			
		Test Seen		Test Unseen		Test Seen		Test Unseen	
		SR	GC	SR	GC	SR	GC	SR	GC
Few-shot (0.5%)	HLSM (Blukis et al. 2021)†	0.82 (N/A)	6.88 (N/A)	0.61 (N/A)	3.72 (N/A)	N/A	N/A	N/A	N/A
	FILM (Min et al. 2022)†	0.00 (N/A)	4.23 (N/A)	0.20 (N/A)	6.71 (N/A)	N/A	N/A	N/A	N/A
	CAPEAM (Kim et al. 2023)	0.00 (0.00)	3.90 (2.29)	0.20 (0.00)	6.63 (2.36)	N/A	N/A	N/A	N/A
	LLM-Planner (Song et al. 2023)	18.20 (N/A)	26.77 (N/A)	16.42 (N/A)	23.37 (N/A)	15.33 (N/A)	24.57 (N/A)	13.41 (N/A)	22.89 (N/A)
	FLARE-LLaMA2	16.96 (4.60)	24.84 (8.09)	17.79 (5.62)	27.40 (9.46)	12.00 (3.01)	20.05 (7.22)	13.73 (4.27)	21.98 (8.46)
	FLARE-Vicuna	20.61 (6.28)	29.57 (10.17)	22.04 (7.61)	33.57 (12.06)	16.37 (4.57)	23.68 (8.84)	18.05 (5.98)	26.75 (10.75)
	FLARE-GPT-3.5	32.55 (12.17)	42.02 (16.94)	31.79 (12.21)	43.94 (17.44)	23.48 (8.71)	33.40 (14.40)	25.38 (9.37)	36.02 (15.28)
	FLARE-GPT-4 (Ours)	**40.05 (16.68)**	**48.84 (21.31)**	**40.88 (18.14)**	**51.72 (22.78)**	**31.96 (12.93)**	**41.36 (18.55)**	**32.57 (12.72)**	**43.23 (18.40)**
Full	HLSM (Blukis et al. 2021)	29.94 (8.74)	41.21 (14.58)	20.27 (5.55)	30.31 (9.99)	25.11 (6.69)	35.79 (11.53)	16.29 (4.34)	27.24 (8.45)
	FILM (Min et al. 2022)	28.83 (11.27)	39.55 (15.59)	27.80 (11.32)	38.52 (15.13)	25.77 (10.39)	36.15 (14.17)	24.46 (9.67)	34.75 (13.13)
	CAPEAM (Kim et al. 2023)	51.79 (21.60)	60.50 (25.88)	46.11 (19.45)	57.33(24.06)	47.36 (19.03)	54.38 (23.78)	43.69 (17.64)	55.66 (22.76)

Table 2: **Planner accuracy comparison.** 'Seen Acc.' and 'Unseen Acc.' denote planner accuracies in valid seen and unseen folds. A plan is regarded as correct when it is aligned with the ground-truth plan. To solely compare planner accuracy, we omit replanning in LLM-Planner and FLARE, denoted with 'Static' and 'w/o EAR'.

LLM	Method	Seen Acc.	Unseen Acc.
LLaMA2	LLM-Planner (Static) (Song et al. 2023)	0.006	0.002
LLaMA2	**FLARE (w/o EAR)**	**18.54**	**22.29**
Vicuna	LLM-Planner (Static) (Song et al. 2023)	8.17	7.06
Vicuna	**FLARE (w/o EAR)**	**24.51**	**33.62**
GPT-3.5	LLM-Planner (Static) (Song et al. 2023)	29.78	31.67
GPT-3.5	**FLARE (w/o EAR)**	**46.10**	**55.66**
GPT-4	LLM-Planner (Static) (Song et al. 2023)	31.54	30.12
GPT-4	**FLARE (w/o EAR)**	**61.34**	**67.48**

Table 3: **Ablation study.** PLW metrics are presented in the parentheses for each metric. MMP and EAR each denotes 'Multi-Modal Planner' and 'Environment Adaptive Replanning,' respectively. We observe that each component contributes to agent's performance.

#	MMP	EAR	Test Seen		Test Unseen	
			SR	GC	SR	GC
(a)	✓	✓	32.55 (12.17)	42.02 (16.94)	31.79 (12.21)	43.94 (17.44)
(b)	✗	✓	30.20 (12.13)	41.26 (17.27)	30.35 (11.62)	42.40 (16.66)
(c)	✓	✗	30.79 (11.98)	40.20 (16.51)	30.28 (12.01)	42.48 (17.03)
(d)	✗	✗	28.05 (11.48)	38.64 (16.23)	28.58 (11.82)	39.92 (16.13)

3.2 seconds (LLM-Planner) to 1.8 seconds while maintaining superior accuracy. The semantic object replacement in EAR completes within 50ms on average, enabling seamless task continuation without noticeable delays in embodied execution.

4.7 Task-Type Analysis and Error Breakdown

To better understand FLARE's performance characteristics, we analyze results across different ALFRED task categories. Our approach shows particularly strong performance on *Look* tasks (48.3% SR) and *Pick* tasks (44.7% SR), where visual grounding provides the most benefit. *Heat* and *Cool* tasks prove more challenging (35.2% and 37.1% SR respectively), often requiring multi-step object interactions that stress our compact representation.

Failure Analysis. We categorize execution failures into: (1) Planning errors (31%) - incorrect subgoal sequences; (2) Navigation failures (28%) - inability to reach target locations; (3) Object manipulation errors (24%) - failed interactions with objects; (4) Semantic replacement failures (17%) - no suitable object alternatives found. EAR successfully handles 68% of object-related failures through semantic replacement, substantially reducing the need for complete task restarts.

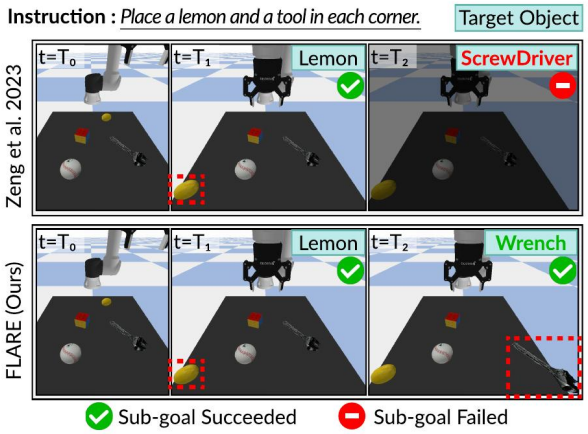

Figure 7: **An example of robotic task applications.** FLARE can be applied to real robotic systems for household task execution.

5 Conclusion

We presented FLARE, a novel approach for few-shot embodied instruction following that combines multi-modal grounded planning with efficient replanning capabilities. Our method addresses key limitations of existing LLM-based planners through two main contributions: (1) a Multi-Modal Planner that retrieves relevant examples using both textual and visual similarities, producing more grounded plans, and (2) Environment Adaptive Replanning that handles execution failures through semantic object replacement without expensive LLM calls.

Experimental results on ALFRED demonstrate the effectiveness of our approach, achieving state-of-the-art few-shot performance with only 0.5% training data. FLARE-GPT-4 obtains 40.88% Test Unseen Success Rate and 51.72% Goal Completion, significantly outperforming existing methods

by +24.46 SR points. Our ablation studies confirm that both MMP and EAR contribute meaningfully to performance, while our compact subgoal representation reduces computational costs by 25%.

The success of FLARE highlights the importance of multi-modal grounding in embodied AI and demonstrates that efficient replanning strategies can enable robust few-shot learning. Our work opens several promising research directions: (1) extending multi-modal retrieval to temporal sequences for learning from demonstration videos; (2) incorporating uncertainty quantification in semantic replacement to improve safety margins; (3) exploring federated learning approaches where multiple agents share their few-shot experiences to accelerate collective learning; and (4) investigating the integration of additional modalities such as audio cues and haptic feedback for richer environmental understanding.

Limitations. While FLARE shows strong performance in simulation, real-world deployment may face challenges with perception accuracy and dynamic environments. Additionally, our approach currently relies on pre-computed object embeddings, which may not generalize to entirely novel object categories.

Broader Impact. Our work advances the development of household robots that can follow natural language instructions with minimal training data. This could lead to more accessible and deployable robotic assistants, potentially improving quality of life for elderly or disabled individuals. However, care must be taken to ensure robust safety mechanisms when deploying such systems in real homes.

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